

Performance Analysis of YOLOv8, YOLOv11, and D-FINE for RoboCup Humanoid KidSize Object Detection

Muhammad Miftah Mahasin Munif, Muchammad Yuda Tri Ananda
Ircham Putra Buwana, Muhammad Mahmud Al-Sawwaf
Diponegoro University, Jl. Prof. Soedarto, Semarang 50275, Indonesia
ewsbascorro@gmail.com

Abstract—This study presents a comparative evaluation of object detection models for RoboCup Humanoid KidSize perception tasks, focusing on the detection of the ball, goal posts, and robots. The evaluated models include YOLOv8, YOLOv11, and D-FINE across multiple scales. Performance is measured using mAP50, mAP50:95, precision, recall, latency, and FPS. Experimental results show that the YOLOv11 family provides the best balance between detection accuracy and real-time performance. In particular, YOLOv11 Large achieves the highest mAP50 of 0.9913, while YOLOv11 Small offers the most efficient real-time performance with 10.11 ms latency and 63.16 FPS. Qualitative analysis also indicates robust detection under multi-object scenes and moderate occlusion. Overall, the results demonstrate that YOLOv11 Small is the most suitable model for real-time humanoid robot perception in RoboCup environments.

Index Terms—RoboCup, humanoid robot, object detection, YOLOv11, real-time perception

I. INTRODUCTION

RoboCup Humanoid KidSize matches impose stringent constraints on onboard perception: limited compute, real-time deadlines, and rapidly changing viewpoints due to locomotion. Object detectors must generalize across diverse arenas and lighting while maintaining sufficiently low latency for closed-loop control. Although deep learning detectors have achieved strong benchmarks in general datasets, their performance in RoboSoccer settings can differ substantially due to domain shift, object scale, and frequent occlusions.

This paper conducts a systematic comparison of three representative detection paradigms: (i) YOLO as a fast one-stage detector; (ii) Faster R-CNN as a canonical two-stage detector with strong accuracy; and (iii) DETR as an end-to-end transformer-based detector. The primary contributions are:

- A task-driven evaluation protocol tailored to RoboCup Humanoid KidSize perception constraints.
- A multi-metric comparison encompassing accuracy, latency, throughput, and resource usage.
- Deployment-oriented discussion and recommendations for practical robot integration.

II. RELATED WORK

A. Object Detection for Mobile and Humanoid Robotics

Object detection is one of the main components in autonomous robotic perception systems, especially in mobile and humanoid robots. In the context of RoboCup Humanoid KidSize, the vision system must be able to recognize important objects such as the ball, goalposts, field lines, and opponent robots under dynamic environmental conditions.

Recent studies show that the application of deep learning in mobile robotics significantly improves detection accuracy and adaptability to environmental changes, particularly under varying lighting conditions and visual noise.

However, the main challenge in humanoid robots lies in limited onboard computational resources and the requirement for real-time inference. In RoboCup systems, detection delays may lead to navigation errors, slower motion responses, and reduced performance during matches.

B. One-Stage Detectors: YOLO

YOLO (You Only Look Once) is a single-stage object detection algorithm that performs classification and bounding box regression in a single inference process. The main advantage of YOLO is its very high inference speed, making it highly suitable for real-time robotic applications.

Research in humanoid soccer robots has shown that YOLO-based models can achieve high speed even on low-specification hardware, reaching approximately 9.66 FPS on Raspberry Pi 3B, which is significantly faster than standard Tiny-YOLO.

In addition, modern versions such as YOLOv5, YOLOv8, and YOLOv9 demonstrate an excellent trade-off between mean Average Precision (mAP) and latency, making them widely used in modern autonomous robotic systems.

However, YOLO may have limitations in small object detection and in detecting objects that are close to each other, which are common situations in RoboSoccer environments.

C. Transformer-Based Detectors: DETR

DETR (Detection Transformer) is a modern approach that utilizes self-attention transformer mechanisms for object detection. Unlike conventional detection methods, DETR does not require region proposal networks or Non-Maximum Suppression (NMS).

This architecture uses a transformer encoder-decoder framework to learn global relationships among visual features, allowing it to better understand the overall context of an image. One of DETR's main advantages is its ability to handle occlusion and spatial relationships between objects, which is highly relevant in robot soccer matches.

However, DETR also has several limitations, including, longer training time, higher computational resource requirements, inference speed that may still be slower than YOLO on edge devices

D. Research Gap

Based on the existing literature, most studies focus on model comparisons in general mobile robotic applications. There is still limited research specifically comparing YOLO and DETR on the RoboCup Humanoid KidSize platform, particularly with emphasis on, real-time latency, small object accuracy, robustness against motion blur, onboard memory usage.

Therefore, this study aims to fill this gap by providing a comprehensive performance analysis of these two detection models.

III. PROBLEM SETTING AND REQUIREMENTS

A. RoboCup Humanoid KidSize Perception Tasks

In the RoboCup Humanoid KidSize competition, robots must be able to detect and track key objects in real-time to effectively perform navigation, control, and game strategy.

For this study, the detection focus is on three main types of objects:

1) *Ball*: The ball is the primary target that must always be detected and tracked. Its position and movement serve as the main input for the robot's control module to determine running direction, dribbling, or kicking actions.

Characteristics: small to medium-sized, often moves quickly, can be occluded by other robots, and arena lighting conditions affect visibility.

2) *Goal*: The goal is an important reference for field orientation and offensive or defensive strategy. The robot must recognize both its own goal and the opponent's goal to evaluate the ball's relative position and make appropriate decisions.

Characteristics: static, larger than the ball, usually has a color that contrasts with the field, but may be affected by light reflections or shadows.

3) *Robot (Opponent or Teammate Robot)*: Other robots must be detected to avoid collisions, predict opponents' trajectories, and coordinate team actions.

Characteristics: dynamically moving, similar in size to the robots used, can occlude each other, and often have similar team-colored uniforms.

4) *Camera Viewpoint*: Cameras are generally mounted on the robot's head, providing a high and limited first-person perspective. Objects can appear at various positions and scales due to robot movement and differences in distance, so the detection system must be able to adapt to object scale and location within the frame.

B. Operational Constraints

Specify constraints such as:

- Onboard compute and memory limits.
- Real-time deadline (e.g., control loop frequency and maximum permissible perception latency).
- Environmental variability (lighting, shadows, reflective floors).
- Motion-induced artifacts (rolling shutter, blur, vibration).

IV. METHODS

A. Detectors Under Comparison

In this study, we evaluate and compare state-of-the-art object detection models from two major architectural paradigms: one-stage convolutional detectors and transformer-based detectors. Specifically, we employ multiple variants of YOLO (You Only Look Once), including YOLOv8 and YOLOv11, alongside the Detection Transformer (DETR) framework.

YOLO-based models are categorized as one-stage detectors, which directly predict object bounding boxes and class probabilities in a single forward pass. This design enables high inference speed and low latency, making them particularly suitable for real-time applications such as humanoid robot soccer. To further investigate the trade-off between computational efficiency and detection performance, we consider multiple model scales, namely nano, small, medium, and large variants. These variants differ in terms of parameter count and computational complexity, allowing a systematic analysis of performance scaling under constrained hardware conditions.

In contrast, DETR represents a fundamentally different approach by leveraging transformer architectures for object detection. Instead of relying on predefined anchor boxes or region proposals, DETR formulates detection as a direct set prediction problem using attention mechanisms. This enables a more global reasoning of object relationships within the scene. However, such advantages often come at the cost of increased computational overhead and longer inference time, which may limit its applicability in real-time robotic systems.

The selected models are chosen to reflect a comprehensive comparison between efficiency-oriented detectors and accuracy-driven transformer-based methods. This comparison is particularly relevant in the context of RoboCup

humanoid soccer, where both detection accuracy and real-time performance are critical. All models are evaluated under consistent experimental conditions to ensure a fair and reliable comparison.

B. Dataset and Annotation Protocol

The dataset used in this study is derived from a publicly available RoboCup humanoid soccer dataset, consisting of a total of 735 images. The dataset focuses on three primary object classes relevant to the RoboCup scenario, namely *ball*, *goal*, and *robot*.

The distribution of annotated objects is imbalanced, with 666 instances of ball, 127 instances of goal, and 1,039 instances of robot. This imbalance reflects the natural occurrence frequency of objects in RoboCup scenarios, where robots appear more frequently than goals, and poses additional challenges for detection models, particularly in accurately recognizing underrepresented classes.

All images are captured in indoor environments under varying lighting conditions. The variability in illumination introduces additional challenges for object detection, as it affects color consistency, contrast, and visibility of objects across different scenes. Such conditions are representative of real-world RoboCup environments.

The dataset annotations are provided in YOLO format (`.txt`), where each object instance is labeled with its corresponding class and normalized bounding box coordinates.

The dataset is divided into training, validation, and testing subsets with a ratio of 82%, 12%, and 6%, respectively, ensuring a balanced evaluation protocol.

Prior to training, all images are resized to a uniform resolution of 512×512 pixels, and automatic orientation correction is applied to maintain consistency in image alignment.

C. Training Setup

All models are trained under a unified experimental configuration to ensure a fair comparison. The YOLO-based models (YOLOv8 and YOLOv11) are trained using the default training settings provided by the Ultralytics implementation, which employs Stochastic Gradient Descent (SGD) as the optimization algorithm. Meanwhile, the DETR model is trained using the AdamW optimizer, following standard practices for transformer-based architectures.

Due to the use of default training configurations, the learning rate scheduling and data augmentation strategies are handled automatically by the respective frameworks. In the case of YOLO, this includes built-in augmentation techniques such as geometric and photometric transformations, which are applied during training to improve generalization performance.

All models are trained for 50 epochs with a batch size of 16. The input image resolution is fixed at 512×512 pixels, consistent with the preprocessing pipeline described in the previous section.

To ensure consistency and comparability, the same dataset splits and preprocessing procedures are applied across all models. This setup allows the evaluation to focus on the inherent differences in model architecture and performance characteristics rather than variations in training configuration.

D. Deployment and Inference Setup

The deployment and inference process is conducted using the same frameworks employed during training to ensure consistency. YOLO-based models are deployed using the Ultralytics framework, while the DETR model is implemented using a standard PyTorch-based inference pipeline.

Inference is performed using default precision settings, without applying additional optimization techniques such as quantization or model acceleration frameworks (e.g., TensorRT). This approach allows the evaluation to reflect the baseline performance of each model without hardware-specific optimizations.

The models are evaluated in a real-time inference scenario, where latency and throughput are critical factors. Inference time is measured in milliseconds per image, and frames per second (FPS) is computed accordingly to assess real-time feasibility.

To maintain fairness, all models are evaluated under consistent conditions using the same input resolution of 512×512 pixels and identical preprocessing steps. This ensures that performance differences are primarily attributed to the architectural characteristics of the models rather than deployment-specific optimizations.

The evaluation focuses on assessing the trade-off between detection accuracy and computational efficiency, which is essential for real-time applications such as humanoid robot soccer.

V. EVALUATION METRICS

We evaluate object detection performance by considering both detection accuracy and real-time feasibility, which are critical for humanoid robot soccer systems operating under dynamic and resource-constrained environments.

- **Detection quality:** Detection performance is evaluated using mean Average Precision (mAP) at different Intersection over Union (IoU) thresholds, including mAP_{50} and $\text{mAP}_{50:95}$. The mAP_{50} metric reflects detection accuracy under a relatively lenient localization constraint, while $\text{mAP}_{50:95}$ provides a stricter and more comprehensive evaluation of localization fidelity across multiple IoU thresholds. In addition, precision and recall are reported to quantify the balance between false positives and false negatives. These metrics are particularly important in scenarios where both accurate object localization and reliable detection are required for downstream decision-making.

- **Runtime:** Real-time performance is assessed using end-to-end inference latency (ms) and throughput measured in frames per second (FPS). Latency represents the processing time per frame, while FPS indicates the system’s ability to sustain real-time operation. In practical robotic systems, a minimum threshold of approximately 30 FPS is often required to ensure smooth perception and control. Higher latency may introduce delays in response, directly affecting navigation, object tracking, and interaction within the environment.
- **Resource usage:** Although detailed measurements such as peak memory usage and power consumption are not explicitly reported, resource efficiency is implicitly evaluated through runtime performance and model scalability. Metrics such as latency and FPS provide indirect insight into computational complexity and suitability for deployment on resource-constrained robotic platforms.
- **Robustness:** Robustness is evaluated in relation to challenging visual conditions such as motion blur, partial occlusion, and illumination variability. These factors are particularly relevant in RoboCup environments, where dynamic movements and non-uniform indoor lighting frequently occur. Given that the dataset includes varying lighting conditions, the evaluation reflects the models’ ability to maintain stable performance under realistic and non-ideal operating scenarios.

VI. EXPERIMENTAL RESULTS

A. Overall Quantitative Results

This study evaluates the detection performance using the mean Average Precision (mAP) metric across several model sizes. The comparison is conducted between YOLO-based models and the D-FINE architecture.

Table I summarizes the performance of all evaluated models across different scales, including detection accuracy and runtime performance.

TABLE I
OVERALL PERFORMANCE COMPARISON ON THE ROBOCUP HUMANOID
KIDSIZE TEST SET.

Method	mAP ₅₀	mAP _{50:95}	Precision	Recall	Latency (ms)	FPS
YOLOv8 (nano)	0.9800	0.8556	0.9555	0.9832	12.05 ms	82.97
YOLOv8 (small)	0.9427	0.7110	0.8543	0.8565	16.59 ms	60.28
YOLOv8 (medium)	0.9890	0.8661	0.9656	0.9854	31.02 ms	32.23
YOLOv8 (large)	0.9850	0.8250	0.9496	0.9654	48.68 ms	20.54
YOLOv11 (nano)	0.9720	0.8563	0.9550	0.9558	12.41 ms	52.02
YOLOv11 (small)	0.9868	0.8715	0.9684	0.9784	10.11 ms	63.16
YOLOv11 (medium)	0.9908	0.8506	0.9769	0.9798	12.62 ms	54.41
YOLOv11 (large)	0.9913	0.8710	0.9867	0.9679	19.25 ms	39.81
D-FINE (nano)	0.9820	0.8190	0.8156	0.9885	26.46 ms	37.79
D-FINE (small)	0.9870	0.8410	0.8665	0.9857	29.24 ms	34.20
D-FINE (medium)	0.9820	0.8480	0.8648	0.9713	37.18 ms	26.90
D-FINE (large)	0.9850	0.8560	0.8912	0.9857	53.54 ms	18.68

Based on the results shown in Table I, YOLOv11 Large records the highest mAP₅₀ of 0.9913, while YOLOv11 Small achieves the best mAP_{50:95} of 0.8715, indicating

strong detection and localization capability across multiple IoU thresholds. The precision and recall values are also consistently high, with YOLOv11 Large reaching 0.9867 precision and 0.9679 recall, and YOLOv11 Small achieving 0.9684 precision and 0.9784 recall.

The YOLOv8 family also shows competitive results, especially the medium variant with mAP₅₀ = 0.9890 and mAP_{50:95} = 0.8661. However, its latency of 31.02 ms is considerably higher than the corresponding YOLOv11 variants, reducing its suitability for real-time deployment.

Similarly, D-FINE remains competitive in terms of accuracy, particularly in the larger variants. For example, D-FINE Large achieves mAP₅₀ = 0.9850 and mAP_{50:95} = 0.8560, but its high latency of 53.54 ms limits its practical use in real-time robotic systems.

Overall, the results indicate that YOLOv11 offers the best balance between detection accuracy and inference efficiency, making it the most suitable choice for real-time humanoid robot perception tasks.

B. Latency and Real-Time Suitability

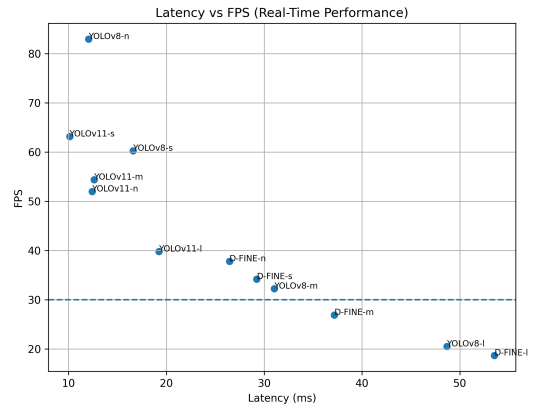


Fig. 1. Latency versus FPS across all evaluated models. The dashed line indicates the 30 FPS threshold for real-time perception.

Figure 1 illustrates the relationship between inference latency and throughput (FPS) for all evaluated models. Models positioned toward the upper-left region of the plot are more desirable, as they achieve both low latency and high frame rates, indicating strong real-time capability.

Latency and real-time suitability are critical in the RoboCup Humanoid KidSize perception pipeline, as the robot must continuously process visual input and respond within a strict control loop. Therefore, latency and FPS must be evaluated jointly to determine whether a model meets real-time operational requirements.

As shown in Fig. 1, most YOLOv11 variants operate well above the 30 FPS threshold, indicating strong real-time performance. In particular, YOLOv11 Small achieves the lowest latency of 10.11 ms while maintaining a high throughput of 63.16 FPS, placing it in the most favorable

region of the plot. Similarly, YOLOv11 Nano and Medium also demonstrate stable real-time capability, consistently exceeding the 30 FPS requirement.

In contrast, several larger models fall closer to or below the real-time threshold. For example, YOLOv8 Large operates at approximately 20.54 FPS, while D-FINE Large achieves only 18.68 FPS, both of which are insufficient for real-time robotic control. Additionally, D-FINE variants generally exhibit higher latency, positioning them in less efficient regions of the plot despite their competitive detection accuracy.

Overall, the results highlight the trade-off between computational complexity and real-time performance. Among all evaluated models, YOLOv11 Small provides the best balance between low latency and high throughput, making it the most suitable choice for real-time humanoid robot perception.

C. Ablation and Sensitivity Studies

The ablation and sensitivity study analyzes how model scale affects both detection accuracy and computational efficiency across YOLOv8, YOLOv11, and D-FINE. In general, increasing the model size improves detection performance but also increases computational cost.

For example, within the YOLOv11 family, mAP50 increases from 0.9720 in the nano variant to 0.9913 in the large variant, while mAP50:95 also improves from 0.8563 to 0.8710. However, this gain is accompanied by higher latency, which rises from 12.41 ms to 19.25 ms, while FPS decreases from 52.02 to 39.81.

A similar trend is observed in D-FINE, but the run-time cost increase is more significant. Although D-FINE Large achieves a competitive mAP50 of 0.9850, its latency reaches 53.54 ms with only 18.68 FPS, indicating higher sensitivity to model scaling compared to the YOLO-based models.

Overall, the results highlight the trade-off between model complexity and deployment efficiency, with YOLOv11 Small providing the most balanced performance for real-time robotic applications.

D. Qualitative Analysis

The qualitative analysis complements the quantitative results by visually evaluating the detection behavior of the models in representative RoboCup Humanoid KidSize scenarios. While metrics such as mAP and recall provide objective performance measures, qualitative observations help assess model behavior in real match environments.

Visual inspection shows that the best-performing models, particularly the YOLOv11 variants, can accurately detect the main objects of interest, namely the ball, goal posts, and robots, with stable bounding box localization. In clear field conditions, the predicted boxes closely align with the actual object boundaries, indicating strong localization consistency.

The models also demonstrate robust detection capability in multi-object scenes, where multiple robots, the ball, and goal posts can be detected simultaneously without significant confidence degradation. In addition, the detector remains relatively robust under challenging conditions such as partial occlusion and small object appearance.

However, several failure cases may still occur, particularly in frames affected by motion blur or visually cluttered scenes, where bounding box precision may decrease and occasional false positives can appear.

VII. DISCUSSION

This study highlights several important trade-offs between detection accuracy and computational performance. One of the main trade-offs observed is between accuracy and latency, where larger models generally achieve higher detection accuracy but require greater computational cost and longer inference time. In contrast, smaller models provide faster inference suitable for real-time applications while still maintaining competitive accuracy, depending on the model architecture.

In terms of robustness, the qualitative results suggest relatively stable performance under moderate occlusion and clutter, such as when multiple robots crowd around the ball. However, detection accuracy may decrease when objects are heavily occluded or only partially visible, especially for small objects.

Another important factor is domain shift, where differences in venues, field textures, and lighting conditions may affect model performance. Models trained on a specific dataset may experience performance degradation when deployed in new environments with different lighting or background conditions.

From a system integration perspective, integration complexity and calibration requirements must also be considered. The detection model must be properly integrated into the robot software stack, including camera calibration, coordinate transformation, and synchronization with other modules such as localization and motion planning. Improper calibration may reduce detection accuracy and overall system performance.

VIII. LIMITATIONS

This study has several limitations. First, the dataset size is relatively limited compared to large-scale object detection datasets, which may affect the model’s generalization capability. Second, the experiments were conducted on a specific GPU platform, so the performance results may vary when implemented on different hardware, such as embedded systems or edge devices.

In addition, there may be bias arising from the training procedure, including dataset distribution, class imbalance, and hyperparameter tuning. The models may perform better on frequently occurring classes, such as robots, compared to less frequent classes, such as goal posts. Hyperparameter tuning was performed within a limited

range, and further optimization may still improve overall performance

IX. CONCLUSION AND FUTURE WORK

In conclusion, this study demonstrates that YOLO-based models provide strong object detection performance for RoboCup humanoid robot applications, offering a good balance between accuracy and real-time performance. In particular, smaller variants such as nano and small models are highly suitable for real-time robotic applications due to their lower latency and higher FPS, while larger variants generally provide slightly higher detection accuracy at the cost of increased inference time.

For future work, several improvements can be explored. These include domain adaptation to improve performance across different environments, temporal models that utilize sequential frame information for more stable detection, and multi-camera fusion to improve detection accuracy and reduce occlusion problems. Additionally, uncertainty-aware detection can be implemented to provide confidence estimation, which is important for safer decision-making in robotic systems.

ACKNOWLEDGMENT

The authors would like to thank all team members and contributors who supported this research and development. We also acknowledge the RoboCup community for providing datasets, evaluation standards, and a competitive environment that encourages continuous improvement in robotic vision systems. This work was further supported by the robotics research team and the laboratory facilities that made this study possible.

REFERENCES

- [1] D. Barry, M. Shah, M. Keijsers, H. Khan, and B. Hopman, "xYOLO: A Model for Real-Time Object Detection in Humanoid Soccer on Low-End Hardware," *arXiv preprint arXiv:1910.03159*, 2019.
- [2] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2017.
- [3] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-End Object Detection with Transformers," *European Conference on Computer Vision (ECCV)*, 2020.
- [4] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, "DETRs Beat YOLOs on Real-time Object Detection," *arXiv preprint arXiv:2304.08069*, 2023.
- [5] Q. Hong, H. Dong, W. Deng, and Y. Ping, "Education Robot Object Detection with a Brain-Inspired Approach Integrating Faster R-CNN, YOLOv3, and Semi-Supervised Learning," *Frontiers in Neurobotics*, 2024.
- [6] J. Terven and D. Cordova-Esparza, "A Comprehensive Review of YOLO Architectures in Computer Vision: From YOLOv1 to YOLOv8 and YOLO-NAS," *arXiv preprint arXiv:2304.00501*, 2023.